

Arnab Bhowmik

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EDUCATION

Stony Brook University

December 2026

Bachelor of Science with Honors in Computer Science

GPA: 3.81

Relevant Coursework: Cloud Computing, Computer Networks, Software Engineering, Theory of Computation: Honors, Analysis of Algorithms: Honors, Machine Learning, Data Structures, Object-Oriented Programming, Systems Programming

EXPERIENCE

CoreWeave

Jun. 2026 – Aug. 2026

Software Engineering Intern, Cloud Platform API/IAM

New York, NY

- Implemented **per-user workload identity** for CoreWeave's **Slurm on Kubernetes (SUNK)**, automatically provisioning short-lived credentials for zero-trust access to cloud services, helping protect **\$10B+** in customer data with no user intervention or long-lived secrets.

Lukacsy-Bhowmik Trading LLC

December 2025 – Present

Co-founder and Quantitative Developer

New York, NY

- Co-founded a quantitative trading firm and designed, built, and deployed an automated market-making system for prediction markets on Polymarket US as one of five firm-wide trading operations.
- Achieved profitability at scale, generating **\$590,000+** in profit (**13,000%+ return** on initial investment).
- Engineered automated quoting, inventory management, and risk controls to continuously provide liquidity across markets, having executed **1.6M+** trades, **129M+** contracts, and over **\$66M** in notional volume.
- Built production infrastructure using Redis, Docker, AWS EC2, and PM2, with automated deployment, monitoring, state management, and alerting for reliable 24/7 operation.

GlaxoSmithKline

May 2025 – Aug. 2025

Compute Platform Engineering Intern

Seattle, WA

- Developed an interactive **Python CLI** that uses workload diagnosis to auto-select optimal HPC environments across **Slurm and Google Batch** and optimize resource specifications when applicable, with two submission modes (automatic job script generation/submission and direct environment access), **reducing compute costs by ~10%**.
- Containerized and deployed the CLI using both **Docker and Apptainer** for cross-platform compatibility on Windows, Linux, and Unix systems, with planned rollout to **3,000+** computational scientists company-wide.
- Built proof of concept demonstrating architectural optimizations for AI/ML team's prototype agentic system's tool orchestration layer, achieving **~35% reduction in context usage**.

Stony Brook University

Jan. 2025 – Dec. 2025

Teaching Assistant

Stony Brook, NY

- Led recitations, review sessions, and office hours for **250+** students across Programming Abstractions (CSE 216) and Software Development (CSE 316).
- Helped revise course materials, graded assignments/exams, and proctored exams to ensure smooth course operations.

Stony Brook University

Sep. 2024 – Present

Student Software Developer, Vertically Integrated Projects (VIP) Program

Stony Brook, NY

- Led the HealthByte subteam and its development by delegating tasks, creating onboarding resources, and organizing meetings.
- Developed prototype patient-facing **iOS** and clinician-facing **full-stack web** applications for post-surgery recovery monitoring using **Swift**, **HealthKit**, and **ResearchKit**, with a centralized **PostgreSQL** database across both platforms.
- Developing **Regio Vinco**, an interactive web-based educational world geography game, iterating on versions tested with **150+** students at partnering schools.

PenguinLearn

Jul. 2024 – May 2025

Full Stack Developer

Remote

- Led full-stack development of **PenguinLearn**, a **RESTful educational platform**, using Next.js, React, Supabase, and Prisma ORM, enabling migration from third-party hosting and **reducing overall operational costs by ~20%**.
- Implemented an on-site **real-time messaging system** for direct communication between parents and teachers, integrated **Stripe payment processing**, and automated class meeting scheduling via the **Zoom API**.
- Built test suites with **Jest and Playwright** and set up a **CI/CD pipeline**, ensuring reliability and streamlined deployments.

PROJECTS

TA Tools | *Python, Flask, Beautiful Soup, Selenium WebDriver, SQLite, JavaScript*

Jul. 2024 – Aug. 2024

- Developed a full-stack web application using Flask, Jinja, and SQLite to **automate logistics tasks** for teaching assistants at a tutoring center, improving task efficiency by approximately **200%** for active users.

SKILLS

Languages: Java, Python, Go, SQL, Bash, C, OCaml, JavaScript, PHP, Swift

Frameworks: NumPy, pandas, React, Vue, LangChain, PyTorch, scikit-learn, Next.js, Express.js, Node.js, FastAPI

Tools: PostgreSQL, MySQL, SQLite, MongoDB, Redis, Pinecone, Git, Docker, Kubernetes, SPIFFE/SPIRE, GitHub Actions, Ansible, Terraform, Slurm, Google Cloud Platform (GCP), Amazon Web Services (AWS)

Concepts: Distributed Systems, RESTful APIs, Agile Development, Machine Learning, Natural Language Processing (NLP), High Performance Computing (HPC), Linux, Unix, LLMs, Model Context Protocol (MCP)